THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE · NCERT-TYPE CHAPTER TEST SET 3

Chapter 2: **Acids, Bases and Salts** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. To dilute concentrated sulphuric acid, one should: 1

(a) add acid slowly to water **(b)** mix them at once
(c) add water to acid **(d)** heat the acid

2. Equal volumes of HCl and acetic acid of the same concentration are taken with magnesium. Bubbles form faster in: 1

(a) HCl **(b)** acetic acid **(c)** both equally **(d)** neither

3. The number of water molecules of crystallisation in blue vitriol is: 1

(a) 10 **(b)** 2 **(c)** 5 **(d)** 7

4. The gas which turns lime water milky is: 1

(a) CO_2 **(b)** H_2 **(c)** N_2 **(d)** O_2

5. The pH of fresh milk is 6. On turning into curd its pH will: 1

(a) increase **(b)** remain 6 **(c)** decrease **(d)** become 7

6. On heating blue crystals of copper sulphate, they become: 1

(a) white **(b)** green **(c)** red **(d)** black

7. The reaction of an acid with a base to give salt and water is called: 1

(a) neutralisation **(b)** displacement **(c)** oxidation **(d)** decomposition

8. The chemical formula of bleaching powder is: 1

(a) $CaCO_3$ **(b)** $CaOCl_2$ **(c)** $CaCl_2$ **(d)** $CaSO_4$

9. The formula of washing soda is: 1

(a) $NaCl$ **(b)** $Na_2CO_3 \cdot 10\,H_2O$ **(c)** Na_2CO_3 **(d)** $NaHCO_3$

10. Gypsum on heating at 373 K gives: 1

(a) washing soda **(b)** plaster of Paris **(c)** bleaching powder **(d)** quick lime

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. What is meant by water of crystallisation? Give one example. 2

12. What is a neutralisation reaction? Give one example. 2

13. Name the acid injected by an ant bite and a remedy. 2

14. Why should curd and sour substances not be kept in brass and copper vessels? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A milkman adds a little baking soda to fresh milk. Why does he shift the pH from 6 to slightly alkaline, and why does it take longer to set as curd? 3

16. Name the products obtained on heating sodium hydrogen carbonate and give three uses of baking soda. 3

17. What is pH? Write the pH of 3
(a) lemon juice
(b) blood
(c) pure water.

OR

Write balanced equations for: 3
(a) NaOH and Zn
(b) HCl and $CaCO_3$
(c) NaOH and CO_2.

18. Classify as acidic, basic or neutral salt solution: NaCl, NH_4Cl, CH_3COONa, Na_2CO_3. 3

OR

Four solutions A, B, C and D have pH 7, 1, 13 and 4 respectively. Classify them and arrange in increasing order of H^+ ion concentration. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Explain neutralisation with three equations. Include one example of an acidic oxide reacting with a base. 5

OR

Explain the role of pH in daily life: acid rain, aquatic life, and self-defence in animals and plants (nettle, bee). 5

20. What is the pH scale? Explain its importance in digestion, tooth decay and soil. 5

OR

Describe acid-base indicators: litmus, phenolphthalein, methyl orange, and olfactory indicators with their colours/odour changes. 5

END OF QUESTION PAPER — Total: 40 Marks

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